W-A-450/17
JANUARY 3,1992

FEDERAL SPECIFICATION

DETECTOR, MOTION, DUAL TECHNOLOGY

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive ~~information and equipment, high dollar items, and weapons~~. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type XVII, Dual Technology Motion Detector.

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The dual technology motion detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

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FSC 6350

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All units shall withstand the tests in 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The dual technology motion detector shall consist of one of the following combinations: passive infrared (PIR) and microwave (MW) motion detectors, or PIR and ultrasonic (US) motion detectors. Usable and published detection patterns shall be included in the manufacturer's literature. Failure to supply this information shall be cause for non-compliance of the sensor. The dual technology sensor shall be designed to stabilize within five minutes after being powered on and to prevent nuisance alarms due to electromagnetic emissions from other equipment such as motors or fluorescent lights (ref: 4.1). Input voltage shall be 8 to 18 volts DC, reverse polarity protected, or 10 to 28 volts AC RMS, 60 cps. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Alarm and tamper switch impedance shall be as follows:

Open state:
- greater than 100 000 ohms.

Closed state:
- For solid state relays - less than 50 ohms.
- For mechanical relays - 0.1 ohms or less.

3.4.1 PIR and MW combined sensor operation.

3.4.1.1 PIR sensor operation. The PIR sensor shall consist of receiver elements and necessary control circuitry to monitor the protected area and to detect changes in infrared radiation within its field-of-view. The sensor shall detect the test target moving in profile to the sensor at a rate of 0.3 m/s anywhere in the sensor's published detection area.

3.4.1.2 MW sensor operation. The MW sensor shall consist of transmitter and receiver elements and necessary control circuitry to saturate the protected area with electromagnetic

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energy. The MW sensor shall operate within a frequency range and power output allocated by the Federal Communications Commission. The sensor shall detect movement anywhere within the published detection area before the test target has moved not more than 1524 mm at any velocity between 0.05 m/s to 3.03 m/s.

3.4.2 PIR and US combined sensor operation.

3.4.2.1 PIR sensor operation. The PIR sensor shall consist of receiver elements and necessary control circuitry to monitor the protected area and to detect changes in infrared radiation within its field-of-view. The sensor shall detect the test target moving in profile to the sensor at a rate of 0.3 m/s anywhere in the sensor's published detection area.

3.4.2.2 US sensor operation. The US sensor shall consist of transmitter and receiver elements and necessary control circuitry to saturate the protected area with acoustical energy at a frequency above 19 kHz. The frequency range may be any frequency between 19 kHz. and 40 kHz. The sensor shall detect movement anywhere within the published detection area before the test target has moved not more than 1500 mm at any velocity between 0.05 m/s to 3.03 m/s.

3.4.3 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature concerning performance data that would exceed the minimum requirements of this specification.

3.5 Dual technology motion detector enclosure. The detector assembly shall be housed in enclosures having a removable cover arranged to provide ready access for mounting, wiring, alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used.

3.6 Field of view alignment. While the dual technology detector is not required to have an adjustable detection pattern, the equipment shall incorporate provision to provide minor adjustments in the aiming of the detection pattern after mounting.

3.7 Cover. The dual technology motion detector covers shall be designed such that any attempted access to the interior shall cause a tamper condition. The tamper circuits shall be in the secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal

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the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.8 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.9 Switch and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the published current rating shall not cause damage to the switch or contacts.

3.10 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.11 Maintainability. The dual technology motion detector shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the sensor beyond mere removal of the cover.

3.12 Life cycle. The detector alarm and tamper switches and/or relays shall be rated of not fewer than 1 000 000 operations under a load of 250 milliampere driven from a 48 volt DC source. Contact resistance shall not change more than +20% from the initial reading.

3.13 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests. The dual technology sensor shall be capable of being tested independently for each technology incorporated into the sensor. A fail-safe circuit shall restore the sensor to operational mode after testing is completed.

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4.1.1 Performance test. The unit being tested shall be mounted in accordance with the manufacturer's recommendations in a room large enough to accommodate 10 percent more than its published detection area. Using a mounted sensor as the origin point, the test engineer shall mark the published detection pattern (PDP) of the IR sensor on the floor of the test area (figure 1). From the origin point of the PDP, radial lines shall be drawn down both of the PDP's side limits to a point 3000 mm beyond the PDP end limit (figure 2). Radials shall be drawn at a 10 degree rotation from these two radials away from the PDP. (figure 3). Additional radials shall be drawn at not more than 10 degree intervals between the two PDP radial limits (figure 4). The test engineer shall also use the origin point of the PDP for the center point of arcs that shall be drawn from one outer radial across the PDP to the other outer radial. The first arc shall have a radius of 1500 mm, the second shall have a radius of 3000 mm, etc. until the end of the radials is reached 3000 mm beyond the PDP end limit (figure 5). A human being not more than 1524 mm tall, about 254 mm wide, and about 36 kg shall be used as a test target for walk testing. The arms of the test target shall be held tightly across the chest to preclude extraneous movements that might affect the accuracy of sensor response data. The room shall be free of all unnecessary infrared, ultrasonic, and microwave sources. The sensor's alarm contacts shall be connected to an a light emitting diode, or an incandescent light bulb to indicate an alarm. If an audio sounder is used, it should be placed in a baffle which absorbs ultrasonic frequencies to prevent interference with the sensor under test. The sensor's sensitivity shall be set at maximum.

4.1.1.1 Performance test with nly PIR technology active. Only the PIR portion of the sensor shall be enabled for this test. The temperature of the test area shall be between 18 degrees C and 33 degrees C. Beginning at the radial outside the detection area limit and along the arc nearest the origin point, the test target shall move at a rate of 0.3 m/s until an alarm is generated. At this point the test target shall stop and mark the location. After the sensor has reset, the test target shall continue to move along the arc until detected. The test target shall again stop and mark the location. This procedure shall continue until the opposite outer radial is reached. The direction the test target moves shall then be reversed, and the same procedure shall be repeated. This two-direction walk test shall be repeated for each arc. A composite of the detection locations shall define the actual detection pattern. The actual detection pattern shall be within plus or minus 5% of the published detection pattern.

4.1.1.2 Performance test with only MW technology active. Only the MW portion of the sensor shall be enabled for this test.

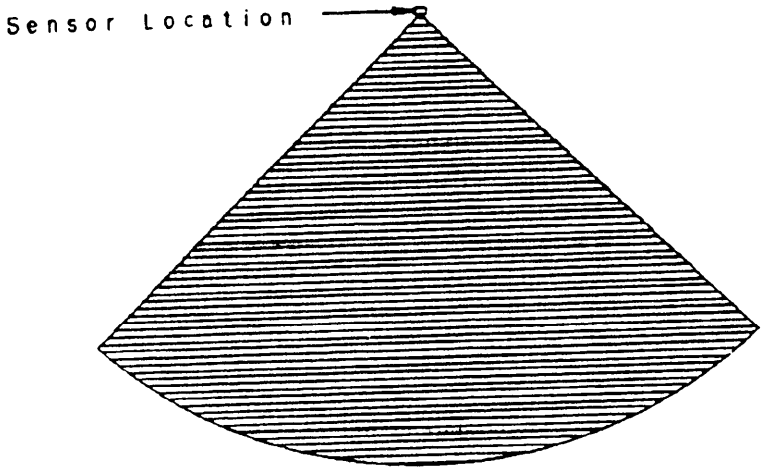
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Beginning at the arc farthest from the origin point and at the end of one of the outer radials, the test target shall approach the sensor straight on at a rate of 0.3 m/s until an alarm is generated. At this point the test target shall stop and mark the location. After the sensor has reset, the test target shall continue moving toward the sensor until detected. This sequence is repeated for the entire length of the radial. This walk test is repeated for all radials. A composite of the marks placed on the floor shall determine the actual detection pattern of the sensor. The actual detection pattern (figure 6) shall be within plus or minus 5% of the published detection pattern.

4.1.1.3 Performance test with only US technology active. Only the US portion of the sensor shall be enabled for this test. Beginning at the arc farthest from the origin point and at the end of one of the outer radials, the test target shall approach the sensor straight on at a rate of 0.3 m/s until an alarm is generated. At this point the test target shall stop and mark the location. After the sensor has reset, the test target shall continue moving toward the sensor until detected. This sequence is repeated for the entire length of the radial. This walk test is repeated for all radials. A composite of the marks placed on the floor shall determine the actual detection pattern of the sensor. The actual detection pattern (figure 6) shall be within plus or minus 5% of the published detection pattern.

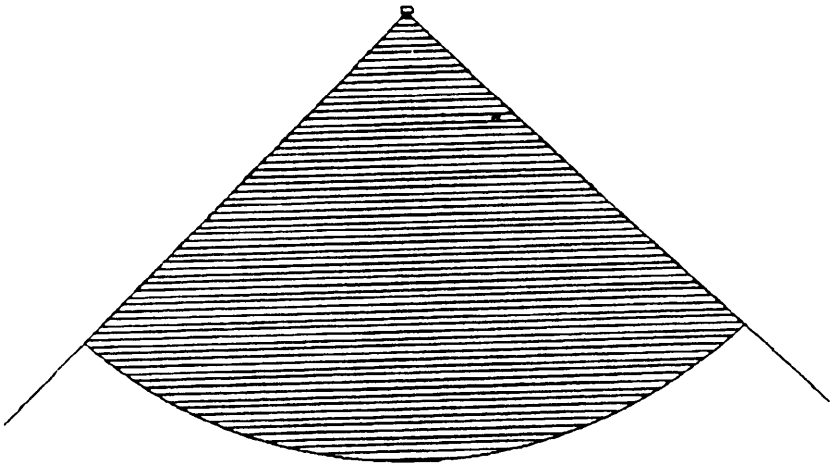
4.1.1.4 Performance test with both technologies active. Some dual technology sensors have the capability to operate with the two technologies in either the "OR" or the "AND" configuration. If the sensor is operated in the "OR" configuration, the actual detection pattern will be a combination of the actual patterns determined in 4.1.1.1 through 4.1.1.2. For this test the sensor shall be operated in the "AND" configuration with both applicable technologies enabled. Repeat the performance testing procedures for the applicable technologies in 4.1.1.1 through 4.1.1.3. The actual usable detection pattern shall be within plus or minus 5% of the published detection pattern.

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Published Detection Pattern of IR Sensor

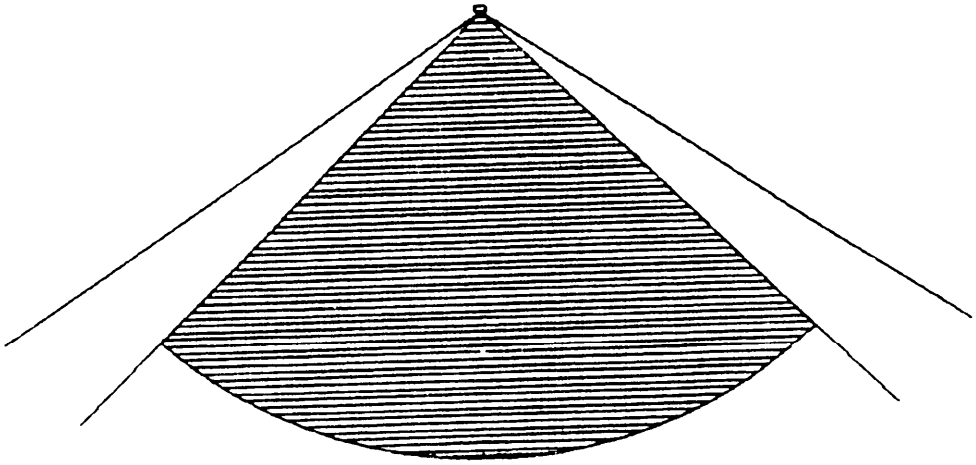
Figure 1



Published Detection Pattern of IR Sensor

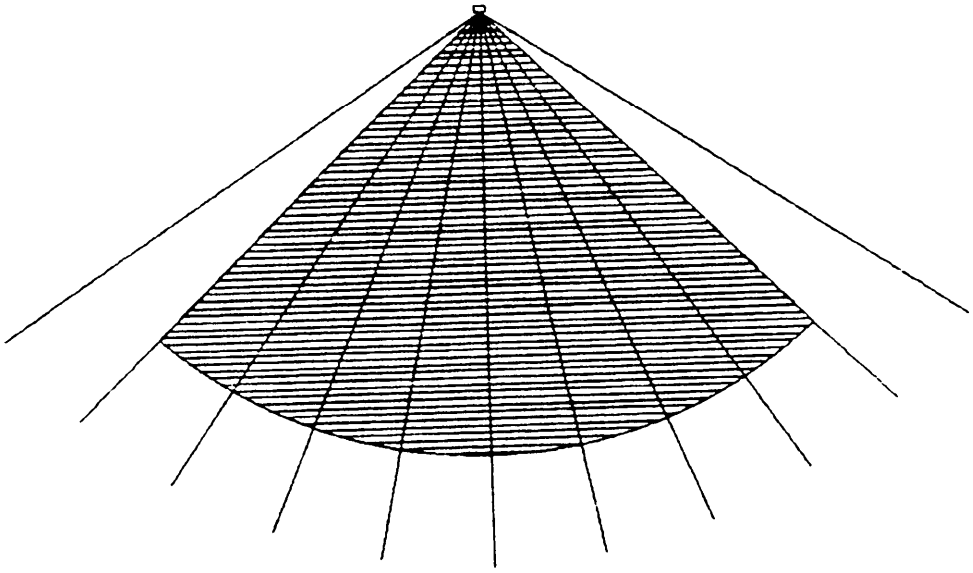
Figure 2

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Published Detection Pattern of IR Sensor

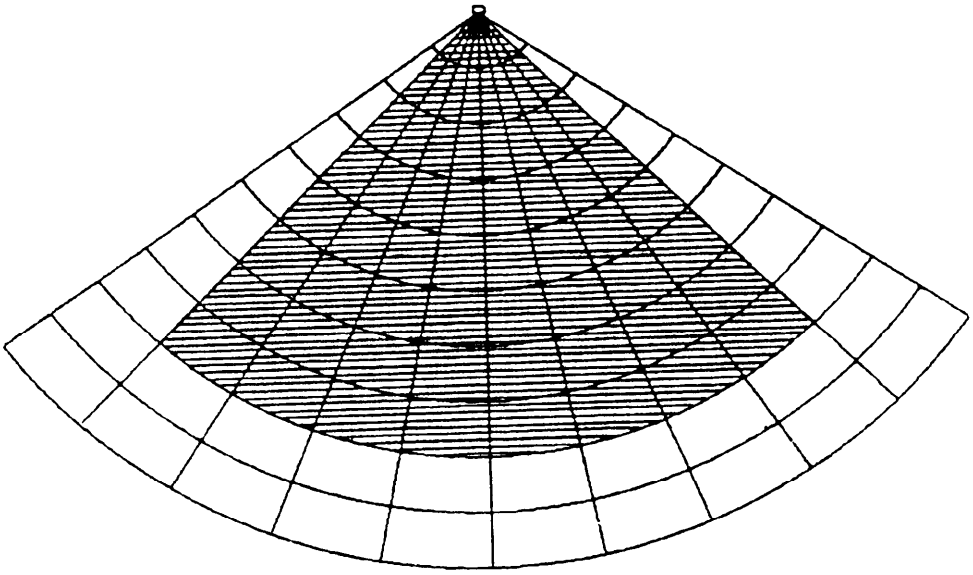
Figure 3



Published Detection Pattern of IR Sensor

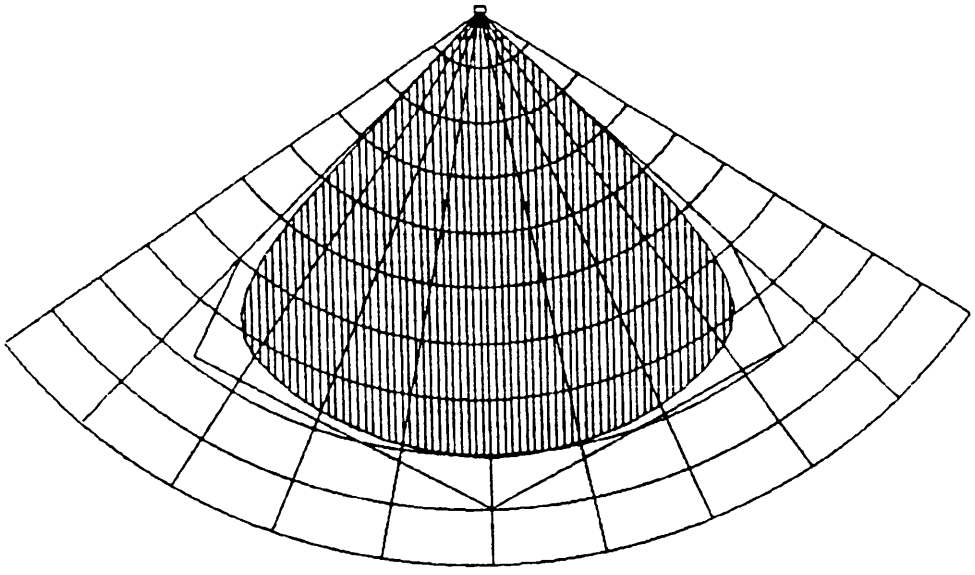
Figure 4

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Published Detection Pattern of IR Sensor

Figure 5



Published Detection Pattern of MW and US Sensors

Figure 6

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4.1.2 Stability test. The equipment submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.13 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

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4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time

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set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.1 Invalid alarm. Any other alarm.

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